

ADVANTAGES AND DISADVANTAGES OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING: A LITERATURE REVIEW

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ABSTRACT

From many years Assisted Instruction (AI) has used to bring the power of the computer system to stand on the educational process. Now a day instead of assisted instruction artificial intelligence methods was applied for the development of intelligent computer assisted instruction (ICAI) systems. It is an attempt to create computerized tutors. This tutors shape the teaching techniques to fit the learning patterns of individual students. This research paper based on the literature reviews available on this emerging topic. The paper gives detail information regarding Artificial Intelligence and machine learning. So that information professionals can use it better way in academic and their research. In this research paper the advantages and disadvantages of artificial intelligence and machine learning are given.

Key words: Artificial intelligence (AI), Machine learning, CAI systems, ICAI systems, Expert System (ES), Natural Language Processing (NLP), Library System, Exploratory Factor Analysis (EFA).

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1. INTRODUCTION

A literature review is a survey of scholarly sources such as books, journal articles, and theses related to a specific topic or recent topics. The purpose of selection of this topic for review is that in today's era application of Artificial Intelligence in library and information centers increases. It is necessary to identify an advantages and disadvantages of AI for the better use of this new technology. Artificial Intelligence has come a long way from its early roots, driven by dedicated researchers. The expression artificial intelligence was introduced as a

‘digital’ replacement for the analog ‘cybernetics’. Artificial intelligence began as an experimental field with pioneers like George Boole (1815-1864), Allen Newell and Herbert Simon, who founded the first artificial intelligence laboratory (Kumar, 2004). Artificial intelligence (AI) is defined as intelligence exhibited by an artificial entity to solve complex problems and such a system is generally assumed to be a computer or machine.

2. DEFINITIONS AND CONCEPTS

2.1. Artificial Intelligence

Artificial intelligence (AI) is an integration of computer science and physiology. Intelligence in simple language is the computational part of the ability to achieve goals in the world. Intelligence is the ability to think to imagine creating memorizing and understanding, recognizing patterns, making choices adapting to change and learn from experience. Artificial intelligence concerned with making computers behave like humans more human like fashion and in much less time than a human takes so is called as Artificial Intelligence (Poole and Goebel, 1998).

2.2. Learning

Learning is defined as the acquiring of knowledge or skill, in a particular domain. This definition is related to human beings. In psychology, various generalized definitions of learning have been proposed, and many of them interpret learning as the change in behavior of a being, subject to a given situation, or as a sequence of his or her repeated experiences in that situation.

2.3. Machine Learning

In simple words learning means either acquiring new knowledge or enhancing or updating individual’s skills. Learning new knowledge is the combination of various processes such as acquisition of significant concepts, understanding their meanings and relationships to each other and to the area concerned. Skill enhancement can be interpreted in biological terms as reinforcing a pattern of neural connections for performing the desired function (Bavakutty, 2006).

Scientific study of algorithms and statistical models when the computer system used to perform a specific task is called Machine learning. It is seen as a subset of artificial intelligence. Machine learning algorithms build a mathematical model based on sample data, known as training data, in order to make predictions or decisions without being explicitly programmed to perform the task (Bishop, 2006).

3. COMPUTER ASSISTED INSTRUCTION SYSTEMS

Computer assisted instruction has been in use for many years, bringing the power of the computer to bear on the educational process. Early CAI programs essentially imitated previously existing instructional materials. For example, a common CAI technique mirrors a method called programmed instruction. In a programmed instruction text, students read brief instructional material and then are presented with short questions to test their comprehension. Students turn to different pages of the book, depending on their answers to the questions, so that each student effectively programs a different path through the material based on individual comprehension abilities. The CAI technique based on programmed instruction is known as frame-based CAI. The effectiveness of frame based CAI program depends entirely on how well it is designed. Although many successful programs have been developed, frame based CAI programs use computers simply as electronic page turners.

An obvious potential application of ES within libraries is for the selection of book sellers or other vendor of library materials carried to its logical conclusion, a system might be developed to select a vendor automate ethical based on past performance in the supply of publications of a particular type such a capability would be especially valuable in the acquisition of material that are less routine-conference proceeding (Mishra and Srivastva, 2008).

4. INTELLIGENT COMPUTER-ASSISTED INSTRUCTION SYSTEMS

Adding AI techniques to CAI results in an effective, new instructional method called Intelligent Computer Assisted Instruction. The goal of ICAI is the development of instructional materials that actually analyze a student's performance in order to develop individualized tutoring strategies. The main components of an ICAI system are problem solving expertise, student model, and tutoring module.

The problem solving expertise component of an ICAI program contains the knowledge that system tries to impart to the student. This expertise is represented with techniques similar to those used in the knowledge base of an expert system. In fact, this component can be thought of as a specialized expert system that contains expertise in domain of the ICAI program. The component of the program that selects the strategies for presenting tutorial information to students is called the tutoring module. A tutoring module generally incorporates natural language processing techniques to communicate at the level of each individual student.

5. AREAS OF ARTIFICIAL INTELLIGENCE

Artificial Intelligence focuses on symbolic, non-algorithmic problem solving methods. Intelligence relies on ability to manipulate symbols. Artificial Intelligence has transformed the society beyond imagination. The goal of its sub areas i.e. expert system, natural language processing, pattern recognition, and robotics is to simulate human intelligence with computers. Some of the recent computational techniques and areas that are utilized in developing fields of Artificial Intelligence are mentioned in Table-1.

Table 1 Areas of Artificial Intelligence

Artificial Intelligence	Cognitive Science Applications	Expert Systems Learning Systems Fuzzy Logic Genetic Algorithms Neural Networks Intelligent Agents
	Robotics Application	Visual Perceptions Tactility Dexterity Locomotion Navigation
	Natural Interface Applications	Natural Languages Speech Recognitions Multisensory Interfaces Virtual Reality

6. APPLICATION OF ARTIFICIAL INTELLIGENCE IN LIBRARIES

Computers provide the perfect medium for the experimentation and application of Artificial Intelligence technology in the libraries. AI has more success at intellectual tasks such as computer based game playing and theorem proving than perceptual tasks. Sometimes these computer programs are intended to stimulate human behavior and they are built for technological applications also such as Computer aided instruction (CAI). In many cases the main goal is to find any technique that does the task quick in the better way. The Table-2 gives the details regarding application of AI and its uses in various areas and activities of libraries.

Table 2 Application of Artificial Intelligence in Libraries

Various Areas of AI		Use and Application
Application of Expert Systems	Library Activities	Dialogue between staff and users Helping in realizing the need for an improvement in the productivity Improve quality of library services Improve quality of information products
	Reference Service	REFSEARCH: Used to teach students reference skills POINTER: Directs the users to the reference sources Online Reference Assistance (ORA) AMSWERMAN: Help users for reference questions on agriculture topics. PLEXUS: Referral tool used in Public Libraries
	Cataloguing	A human-machine interface, where the intellect effort is divided between the intermediary and the support system; An Expert System with full cataloguing capability linked into electronic publishing system, so that as a text is generated on-line.
	Classification	Coal SORT: Conceptual browser designed to serve either as a search or an indexing tool. EP-X: The Environmental Pollution Expert(EP-X) has certain things in common with coal SORT BIOSIS: Uses a knowledgebase to assign documents to categories automatically.
	Indexing	Med Index
	Acquisitions	A subject bibliographer makes in selecting monographic
Application of Natural Language Processing	Library Activities	To apply this is to more specifically to searching database such as online public access catalogs (OPAC) Use of natural language for Dialog database searches
Application of Pattern Recognitions	Library Activities	Use in digital libraries, multimedia systems, geographic information systems, and collaborative computing to electronic commerce have created tremendous opportunities for information researchers. Focus on the user-friendly, graphical, & seamless Human-Computer Interactions.
Application of Robotics	Library Activities	Comprehensive Access to Printed Material (CAPM) is to build a robotic, on-demand and batch scanning system that will allow for real-time browsing of printed material through a web interface.

7. ADVANTAGES AND DISADVANTAGES OF ARTIFICIAL INTELLIGENCE (AI)

One of the major advantages of artificial intelligence is that its decisions are based on facts rather than emotions. Even after our utmost efforts, it is a well known fact that human decisions are always affected in a negative way by our emotions.

7.1. Advantages of Artificial Intelligence (AI)

The benefits of artificial intelligence are incredible, what this area can offer us, is to evolve definitively and move on to the history of artificial robots. Following are the main advantages of Artificial Intelligence (AI).

- Finished task faster than a human,
- Stressful and complex work completed easily,
- Difficult work done in short period,
- Various functions can done at a time,
- Success ratio is high,
- Less errors in task and defects also,
- More efficiency in short time,
- Less space, less size,
- Calculation of long term and complex situations, and
- Discover unexplored things. i.e. outer space.

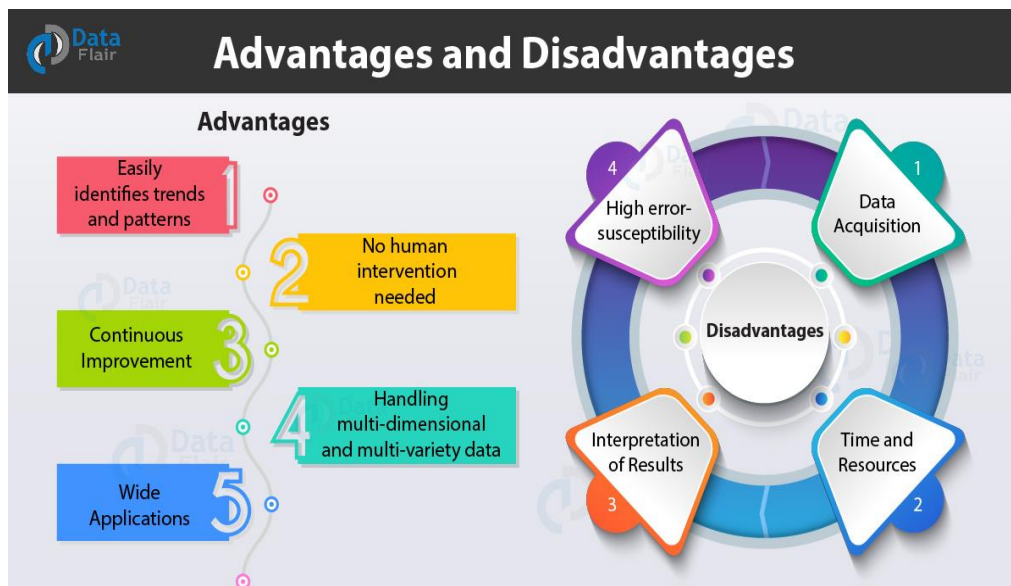
7.2. Disadvantages of Artificial Intelligence (AI)

Some of the main disadvantages of Artificial Intelligence (AI) in our daily lives are as follows.

- Some time it can be misused leading to mass scale destruction,
- Programme mismatch sometime done opposite to the command,
- Human jobs affected,
- Unemployment problem increased,
- Creativity is depend upon programmer,
- Lacks the human touch,
- Younger generation becomes lazy,
- Require a lot of time and money, and
- Technological dependency increased.

8. ADVANTAGES AND DISADVANTAGES OF MACHINE LEARNING LANGUAGE

Machine learning not only does it offer a remunerative career, it promises to solve problems and also benefit companies and non-profit organizations by making predictions and helping them make better decisions. To know clearly where it will be used and where not, it is necessary to identify advantages and disadvantages of machine learning. The advantages and disadvantages of machine learning languages are shown in the following figure.



(Source: <https://data-flair.training/blogs/advantages-and-disadvantages-of-machine-learning>)

9. CONCLUSIONS

In the future AI will enable new capabilities to address library user's information needs. Libraries can use AI tools to provide not just information but deep intelligence offering Insight As A Service (IAAS). It is necessary to understand the advantages and disadvantages of Artificial Intelligence and machine learning for better user and its application in Libraries and Information Centers. AI is impacting the way information is processed and searched for and information professionals will be able to use these exciting new technologies to enhance their services and help users find and access specific information more easily and quickly. Library and Information centers will get benefits by the development of the efficient expert system for technical services as well as Information processing and management.

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